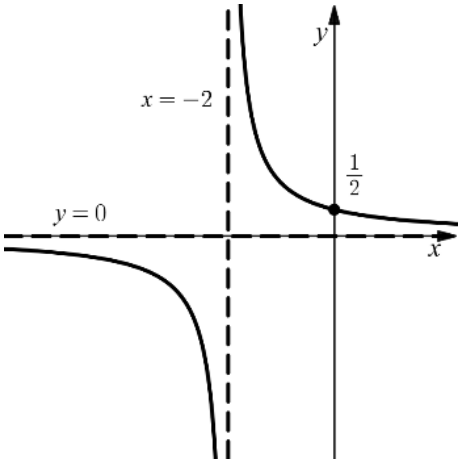
1) a)

Asymptotes: $x = -2, y = 0$

When $x = 0, y = \frac{1}{2}$



$\{x: x \in \mathbb{R}, x < -2\}$

2) a)

$x \in \left[a, -\frac{1}{2}\right) \cup (1, b]$

2) b)

$x \in (-1, a] \cup \left[b, \frac{5}{2}\right)$

2) c)

$x \in [a, -1) \cup \left(\frac{2}{3}, 3\right)$

3) a)

$\frac{1}{x^2-4} = \frac{1}{(x-2)(x+2)}$

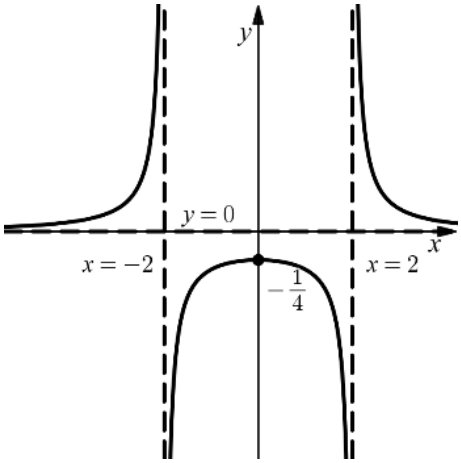
Asymptotes: $x = -2$, $x = 2$ and $y = 0$

Min at $(0, -4)$ becomes local max at $\left(0, -\frac{1}{4}\right)$

As $x \to -\infty$, $\frac{1}{\mathrm{f}(x)} \to 0^+$, as $x \to \infty$, $\frac{1}{\mathrm{f}(x)} \to 0^+$

As $x \to -2$ from LHS $\frac{1}{\mathrm{f}(x)} \to \infty$, as $x \to -2$ from RHS $\frac{1}{\mathrm{f}(x)} \to -\infty$

As $x \to 2$ from LHS $\frac{1}{\mathrm{f}(x)} \to -\infty$, as $x \to 2$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$



$x \in (-2, 2)$

3) b)

$\frac{1}{6+5x-x^2} = \frac{1}{-(x+1)(x-6)}$

When $x = 0$, $y = \frac{1}{6}$

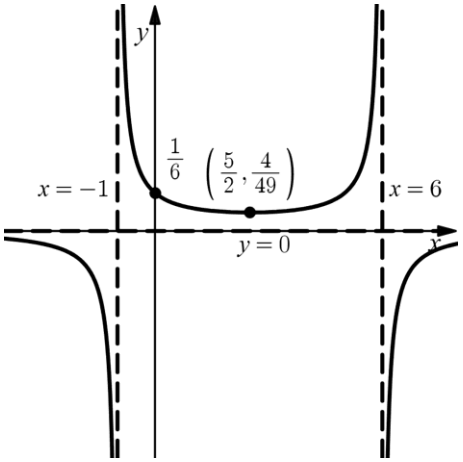
Asymptotes: $x = -1$, $x = 6$ and $y = 0$

Max at $\left(\frac{5}{2}, \frac{49}{4}\right)$ becomes local max at $\left(\frac{5}{2}, \frac{4}{49}\right)$

As $x \to -\infty$, $\frac{1}{\mathrm{f}(x)} \to 0^-$, as $x \to \infty$, $\frac{1}{\mathrm{f}(x)} \to 0^-$

As $x \to -1$ from LHS $\frac{1}{\mathrm{f}(x)} \to -\infty$, as $x \to -1$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$

As $x \to 6$ from LHS $\frac{1}{\mathrm{f}(x)} \to \infty$, as $x \to 6$ from RHS $\frac{1}{\mathrm{f}(x)} \to -\infty$



$x \in (-\infty, -1) \cup (6, \infty)$

4) a)

$\frac{1}{x^2-9} = \frac{1}{(x-3)(x+3)}$

When $x = 0$, $y = -\frac{1}{9}$

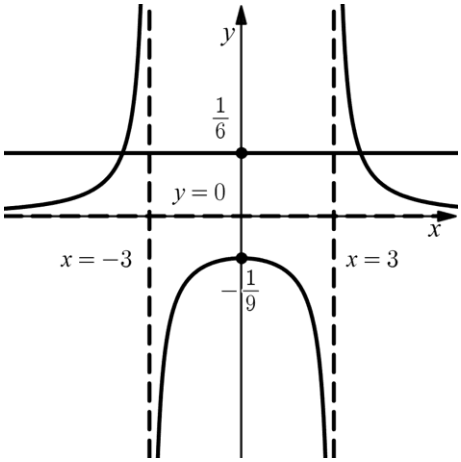
Asymptotes: $x = -3$, $x = 3$ and $y = 0$

Min at $(0, -9)$ becomes local max at $\left(0, -\frac{1}{9}\right)$

As $x \to -\infty$, $\frac{1}{\mathrm{f}(x)} \to 0^+$, as $x \to \infty$, $\frac{1}{\mathrm{f}(x)} \to 0^+$

As $x \to -3$ from LHS $\frac{1}{\mathrm{f}(x)} \to \infty$, as $x \to -3$ from RHS $\frac{1}{\mathrm{f}(x)} \to -\infty$

As $x \to 3$ from LHS $\frac{1}{\mathrm{f}(x)} \to -\infty$, as $x \to 3$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$



$\frac{1}{6} = \frac{1}{x^2-9} \Rightarrow x^2 - 9 = 6 \Rightarrow x^2 - 15 = 0 \therefore x = \pm\sqrt{15}$

$\{x: x \in \mathbb{R}, -\sqrt{15} < x < -3\} \cup \{x: x \in \mathbb{R}, 3 < x < \sqrt{15}\}$

4) b)

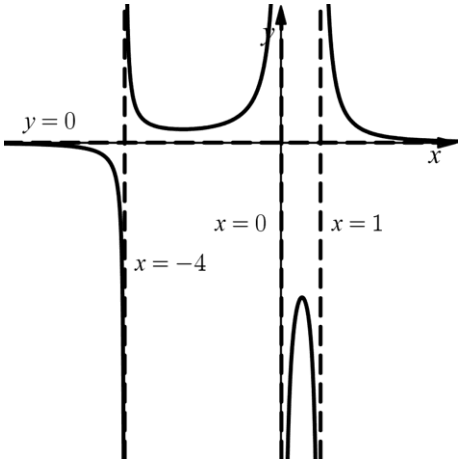
Asymptotes: $x = -4$, $x = 0$ and $x = 1$

As $x \to -\infty$, $\frac{1}{\mathrm{f}(x)} \to 0^-$, as $x \to \infty$, $\frac{1}{\mathrm{f}(x)} \to 0^+$

As $x \to -4$ from LHS $\frac{1}{\mathrm{f}(x)} \to -\infty$, as $x \to -4$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$

As $x \to 0$ from LHS $\frac{1}{\mathrm{f}(x)} \to \infty$, as $x \to 0$ from RHS $\frac{1}{\mathrm{f}(x)} \to -\infty$

As $x \to 1$ from LHS $\frac{1}{\mathrm{f}(x)} \to -\infty$, as $x \to 1$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$



$\{x: x \in \mathbb{R}, x < -4\} \cup \{x: x \in \mathbb{R}, 0 < x < 1\}$

5) a)

$\mathrm{f}\left(\frac{1}{4}\right) = 12\left(\frac{1}{4}\right)^3 + \left(\frac{1}{4}\right)^2 - 9\left(\frac{1}{4}\right) + 2 = \frac{3}{16} + \frac{1}{16} - \frac{9}{4} + 2 = 0$

$\therefore$ $(4x - 1)$ is a factor of $\mathrm{f}(x) = 12x^3 + x^2 - 9x + 2$

5) b)

$12x^3 + x^2 - 9x + 2 = (4x - 1)(3x^2 + x - 2) = (4x - 1)(3x - 2)(x + 1)$

$\frac{1}{12x^3+x^2-9x+2} = \frac{1}{(4x-1)(3x-2)(x+1)}$

When $x = 0$, $y = \frac{1}{2}$

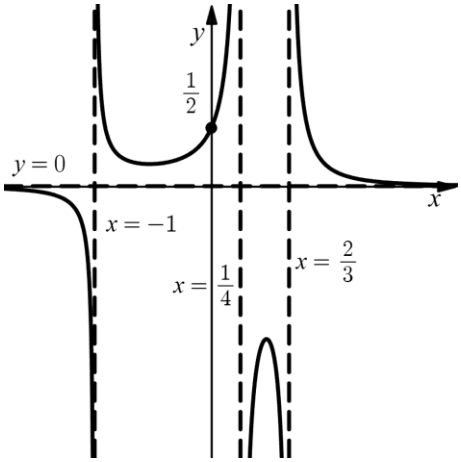
Asymptotes: $x = -1$, $x = \frac{1}{4}$, $x = \frac{2}{3}$ and $y = 0$

As $x \to -\infty$, $\frac{1}{\mathrm{f}(x)} \to 0^-$, as $x \to \infty$, $\frac{1}{\mathrm{f}(x)} \to 0^+$

As $x \to -1$ from LHS $\frac{1}{\mathrm{f}(x)} \to -\infty$, as $x \to -1$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$

As $x \to \frac{1}{4}$ from LHS $\frac{1}{\mathrm{f}(x)} \to \infty$, as $x \to \frac{1}{4}$ from RHS $\frac{1}{\mathrm{f}(x)} \to -\infty$

As $x \to \frac{2}{3}$ from LHS $\frac{1}{\mathrm{f}(x)} \to -\infty$, as $x \to \frac{2}{3}$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$



5) c)

$\{x: x \in \mathbb{R}, x < -1\} \cup \left\{x: x \in \mathbb{R}, \frac{1}{4} < x < \frac{2}{3}\right\}$

6) a)

$\mathrm{f}\left(\frac{1}{3}\right) = 6\left(\frac{1}{3}\right)^3 - 5\left(\frac{1}{3}\right)^2 - 2\left(\frac{1}{3}\right) + 1 = \frac{2}{9} - \frac{5}{9} - \frac{2}{3} + 1 = 0$

$\therefore$ $(3x - 1)$ is a factor of $\mathrm{f}(x) = 6x^3 - 5x^2 - 2x + 1$

6) b)

$6x^3 - 5x^2 - 2x + 1 = (3x - 1)(2x^2 - x - 1) = (3x - 1)(2x + 1)(x - 1)$

$\frac{1}{6x^3-5x^2-2x+1} = \frac{1}{(3x-1)(2x+1)(x-1)}$

When $x = 0$, $y = 1$

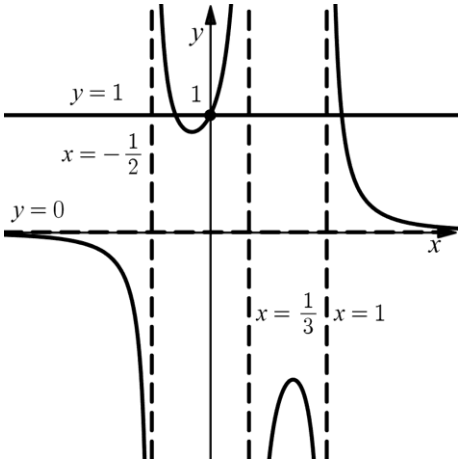
Asymptotes: $x = -\frac{1}{2}$, $x = \frac{1}{3}$, $x = 1$ and $y = 0$

As $x \to -\infty$, $\frac{1}{\mathrm{f}(x)} \to 0^-$, as $x \to \infty$, $\frac{1}{\mathrm{f}(x)} \to 0^+$

As $x \to -\frac{1}{2}$ from LHS $\frac{1}{\mathrm{f}(x)} \to -\infty$, as $x \to -\frac{1}{2}$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$

As $x \to \frac{1}{3}$ from LHS $\frac{1}{\mathrm{f}(x)} \to \infty$, as $x \to \frac{1}{3}$ from RHS $\frac{1}{\mathrm{f}(x)} \to -\infty$

As $x \to 1$ from LHS $\frac{1}{\mathrm{f}(x)} \to -\infty$, as $x \to 1$ from RHS $\frac{1}{\mathrm{f}(x)} \to \infty$



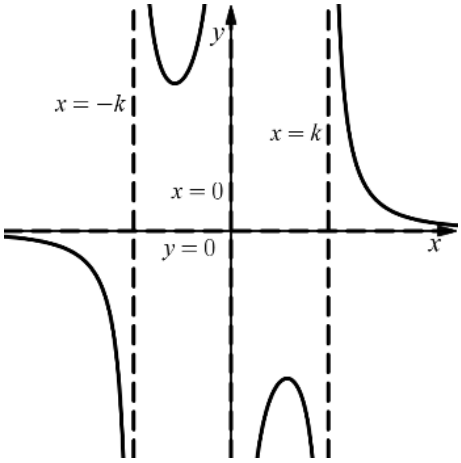
6) c)

$\frac{1}{6x^3-5x^2-2x+1} = 1 \Rightarrow 1 = 6x^3 - 5x^2 - 2x + 1 \Rightarrow 6x^3 - 5x^2 - 2x = 0$

$\Rightarrow x(6x^2 - 5x - 2) = 0 \therefore x = 0, x = \frac{5}{12} \pm \frac{\sqrt{73}}{12}$

$x \in \left(-\infty, -\frac{1}{2}\right) \cup \left(\frac{5}{12} - \frac{\sqrt{73}}{12}, 0\right) \cup \left(\frac{1}{3}, 1\right) \cup \left(\frac{5}{12} + \frac{\sqrt{73}}{12}, \infty\right)$

7) a)



7) b)

$\{x: x \in \mathbb{R}, -k < x < 0\} \cup \{x: x \in \mathbb{R}, x > k\}$